Initial Project Planning

Date	29 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Team Members	Chikali Sarika, Anusha Gollapalli, Karishma Sharon Kaparapu, Ambika Chitikela, Parneeth Devanaboina
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product backlog and sprint schedule for the HDI Predictor project, spanning four sprints across data preparation, model building, web app development, and deployment.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I can collect historical country HDI indicators (life expectancy, schooling years, GNI per capita) from UNDP/Kaggle datasets so that the model has quality training data.	3	High	Chikali Sarika	26-June-2026	26-June-2026
Sprint-1		USN-2	As a developer, I can clean and preprocess the dataset (handle missing values, normalize indicators) so that the model receives consistent inputs.	3	High	Anusha Gollapalli	26-June-2026	26-June-2026
Sprint-1	Exploratory Data Analysis	USN-3	As a developer, I can visualize indicator distributions and correlations using Matplotlib/Seaborn to understand patterns in the data.	2	Medium	Karishma Sharon Kaparapu	26-June-2026	26-June-2026
Sprint-2	Model Building	USN-4	As a developer, I can train a classification model using scikit-learn to predict a country's HDI tier from its indicators.	5	High	Ambika Chitikela	27-June-2026	27-June-2026
Sprint-2		USN-5	As a developer, I can evaluate model performance using accuracy, precision, recall, and F1-score to ensure reliable predictions.	3	High	Parneeth Devanaboina	27-June-2026	27-June-2026
Sprint-2		USN-6	As a developer, I can tune model hyperparameters to improve classification performance across all HDI tiers.	3	Medium	Chikali Sarika	27-June-2026	27-June-2026
Sprint-2	Model Persistence	USN-7	As a developer, I can save the trained model as a pickle (.pkl) file so it can be loaded and reused by the web application.	1	High	Anusha Gollapalli	28-June-2026	28-June-2026
Sprint-3	Web Application	USN-8	As a user, I can enter life expectancy, schooling years, and GNI per capita through a web form so that I can request an HDI prediction.	3	High	Karishma Sharon Kaparapu	29-June-2026	29-June-2026

Sprint-3		USN-9	As a user, I can view the predicted HDI tier (Very High/High/Medium/Low) instantly after submitting my indicator values.	2	High	Ambika Chitikela	29-June-2026	30-June-2026
Sprint-3		USN-10	As a user, I can view a chart showing which indicators most influenced my predicted HDI tier.	2	Medium	Parneeth Devanaboina	30-June-2026	30-June-2026
Sprint-4	Testing & Deployment	USN-11	As a developer, I can test the Flask application across the Very High, Medium, and Low HDI scenarios to confirm correct behavior.	2	High	Chikali Sarika	30-June-2026	30-June-2026
Sprint-4		USN-12	As a developer, I can deploy the Flask application so that it is accessible to policymakers and researchers.	2	Medium	Anusha Gollapalli	1-July-2026	1-July-2026
Sprint-4	Documentation	USN-13	As a team, we can document the project architecture, setup, and usage instructions for future reference.	2	Low	Ambika Chitikela	1-July-2026	1-July-2026